Wi-Fi & Network Troubleshooting Guide

Common problems, step-by-step fixes, commands and a diagnostic flowchart

Covers	10 common Wi-Fi/LAN problems, command cheat sheet, diagnostic flowchart, router basics
Tools	Windows CMD, Python 3, Cisco Packet Tracer (optional)
Author	Krishnakumar C, Chennai

1. Introduction

This handbook covers the most common Wi-Fi and LAN problems faced by home users and small offices, with a repeatable way to find and fix each one. It is written for desktop support and network support work.

Troubleshooting method

1. Identify the problem: what exactly is not working, for whom, and since when?
2. Check the physical layer first: cables, link lights, Wi-Fi switch, router power.
3. Isolate: one device or all devices? One website or all websites?
4. Test with commands (ping, ipconfig, nslookup) and go layer by layer.
5. Fix one thing at a time, then test again.
6. Verify the fix and document what you did.

2. Network basics

Term	Meaning
IP address	Unique address of a device on the network (e.g. 192.168.1.10).
Subnet mask	Tells which part of the IP is the network and which is the host (e.g. 255.255.255.0).
Default gateway	The router's IP; the way out of the local network.
DNS	Converts names (google.com) into IP addresses.
DHCP	Automatically gives devices an IP, mask, gateway and DNS.
APIPA (169.254.x.x)	Self-assigned address when no DHCP server answers.
TCP vs UDP	TCP is reliable and connection-based (web, email). UDP is faster, no guarantee (video calls, DNS queries).
2.4 GHz vs 5 GHz	2.4 GHz has longer range but is slower and crowded. 5 GHz is faster but has shorter range.

OSI layers (troubleshooting view)

Layer	Examples	Typical problems
7 Application	HTTP, DNS, DHCP	Website or DNS errors
4 Transport	TCP, UDP	Ports blocked by firewall
3 Network	IP, ICMP (ping)	Wrong IP, gateway, routing
2 Data Link	Ethernet, Wi-Fi, MAC	Wi-Fi not connecting, switch port
1 Physical	Cables, radio signal	Cable fault, weak signal

3. Problems and fixes

Each problem follows the same layout: Symptom, Possible causes, Steps to check, Fix, Commands.

Problem 1: No internet, but Wi-Fi shows connected

Symptom: Wi-Fi icon shows connected but websites do not open.

Possible causes

- ISP or modem outage
- Router lost its WAN connection
- DNS failure
- Wrong or missing gateway

Steps to check

1. Check if other devices on the same Wi-Fi also have no internet.
2. Ping the gateway (router). If it fails, the problem is the local connection.
3. Ping 8.8.8.8. If it works but names fail, it is a DNS problem (see Problem 5).
4. If 8.8.8.8 fails but the gateway works, the problem is the router/ISP side.

Fix

- Restart modem and router (off 30 seconds, then on).
- Check the WAN/Internet light on the router.
- Call the ISP if the WAN light is red or off.

Commands

```
ping 192.168.1.1
ping 8.8.8.8
ping google.com
ipconfig /all
```

Problem 2: Cannot connect to Wi-Fi

Symptom: Network is visible but the device will not connect, or keeps asking for the password.

Possible causes

- Wrong password
- Saved profile is corrupted
- Band mismatch (2.4/5 GHz)
- MAC filtering on the router
- Router has reached its device limit

Steps to check

1. Re-enter the password carefully (check Caps Lock).
2. Forget the network and connect again.
3. Try the other band (2.4 GHz or 5 GHz).
4. Check if the router has MAC filtering or a device limit.
5. Test with another device to see if the issue is the router or the device.

Fix

- Forget and reconnect.
- Update or reinstall the Wi-Fi adapter driver.
- Restart the router.
- Remove MAC filter entry or raise the device limit.

Commands

```
netsh wlan show profiles
netsh wlan delete profile name="WiFiName"
netsh wlan show interfaces
```

Problem 3: Limited connectivity / 169.254.x.x IP (APIPA)

Symptom: 'Limited or no connectivity' message; ipconfig shows an address starting with 169.254.

Possible causes

- DHCP server (router) not responding
- DHCP disabled on router
- Bad cable or weak Wi-Fi signal
- Adapter problem

Steps to check

1. Run ipconfig /all and confirm the 169.254 address.
2. Check the cable or Wi-Fi signal.
3. Release and renew the IP.
4. Log in to the router and confirm DHCP is enabled and has free addresses.

Fix

- ipconfig /release then ipconfig /renew.
- Restart the router.
- Enable DHCP on the router, or set a static IP temporarily.

Commands

```
ipconfig /all
ipconfig /release
ipconfig /renew
```

Problem 4: IP address conflict

Symptom: 'Another device on the network has the same IP address'; connection drops now and then.

Possible causes

- Two devices with the same static IP
- A static IP inside the DHCP range
- Old DHCP lease

Steps to check

1. Note the IP address shown in the error.
2. Run arp -a to see which MAC addresses are using it.
3. Identify the other device and check whether it uses a static IP.

Fix

- Set the PC to obtain an IP automatically.
- Give static devices addresses outside the DHCP range.
- Release and renew, or use DHCP reservation on the router.

Commands

```
arp -a
ipconfig /all
ipconfig /release
ipconfig /renew
```

Problem 5: DNS error: sites do not open but ping works

Symptom: 'DNS_PROBE_FINISHED_NO_INTERNET' or 'server not found'; ping 8.8.8.8 works but ping google.com fails.

Possible causes

- ISP DNS server down
- Wrong DNS settings
- Corrupt DNS cache

Steps to check

1. Ping 8.8.8.8 (works) and ping google.com (fails): confirms DNS.
2. Run nslookup google.com and see if a server answers.
3. Check which DNS server the PC is using in ipconfig /all.

Fix

- ipconfig /flushdns.
- Set DNS manually to 8.8.8.8 and 1.1.1.1.
- Restart the router.

Commands

```
nslookup google.com
ipconfig /flushdns
ipconfig /all
```

Problem 6: Slow internet

Symptom: Pages load slowly, videos buffer, downloads are slow.

Possible causes

- Weak Wi-Fi signal
- Too many devices connected
- Channel interference
- Background downloads or updates
- ISP speed problem

Steps to check

1. Run a speed test on a wired and a Wi-Fi device and compare.
2. Check signal strength and distance from the router.
3. See how many devices are connected and what is using bandwidth.
4. Look at the Wi-Fi channel; crowded channels cause interference.

Fix

- Move closer to the router or use 5 GHz.
- Change the Wi-Fi channel (1, 6 or 11 on 2.4 GHz).
- Disconnect unused devices.
- Restart the router; contact the ISP if the wired speed is also low.

Commands

```
ping -t 8.8.8.8 (watch time=)
netsh wlan show interfaces (signal %)
netstat -e
```

Problem 7: Cannot reach the default gateway

Symptom: Ping to the router fails or times out; no access to the router page.

Possible causes

- Wrong gateway setting
- Wrong subnet mask
- Cable/port fault
- Router down

Steps to check

1. Run ipconfig /all and compare the IP, mask and gateway: they must be in the same network.
2. Ping the gateway; if it fails, check the cable or Wi-Fi.
3. Run tracert 8.8.8.8 and see where the first hop fails.

Fix

- Switch the adapter back to automatic (DHCP) settings.
- Correct the static gateway or subnet mask.
- Replace the cable or try another router port.

Commands

```
ipconfig /all
ping <gateway IP>
tracert 8.8.8.8
route print
```

Problem 8: Network cable or switch port not working

Symptom: No link light, 'Network cable unplugged' message, or connection that keeps dropping.

Possible causes

- Damaged or loose cable
- Faulty port on switch/router
- Faulty network card
- Duplex or speed mismatch

Steps to check

1. Check that the cable is firmly seated at both ends and the link lights are on.
2. Swap the cable with a known good one.
3. Try a different port on the switch or router.
4. Test the cable on another PC to isolate PC versus cable.

Fix

- Replace the cable or re-crimp the connector.
- Move to a working port.
- Update or reinstall the NIC driver.
- Reset adapter speed/duplex to Auto.

Commands

```
ipconfig /all
netsh interface show interface
Device Manager > Network adapters
```

Problem 9: Printer or shared folder not reachable on the LAN

Symptom: Cannot print to a network printer or open \\PC-NAME shared folders.

Possible causes

- Devices on different subnets
- Firewall blocking file/printer sharing
- Network profile set to Public
- Printer IP changed

Steps to check

1. Ping the printer or PC by IP address.
2. If the IP works but the name does not, it is a name resolution problem.
3. Check that the network profile is Private and file/printer sharing is on.
4. Check whether the printer got a new IP from DHCP.

Fix

- Set the network profile to Private.
- Allow File and Printer Sharing in the firewall.
- Give the printer a static IP or a DHCP reservation.
- Re-add the printer using its IP address.

Commands

```
ping <printer IP>
ping <PC-NAME>
net view
arp -a
```

Problem 10: VPN not connecting

Symptom: VPN client shows 'connecting' forever or fails with an error.

Possible causes

- No base internet connection
- Wrong credentials or expired password
- Firewall or ISP blocking VPN ports
- Outdated VPN client

Steps to check

1. Confirm normal internet works first (ping 8.8.8.8).
2. Check username, password and server address.
3. Try another network (for example a mobile hotspot) to see if the current network blocks the VPN.
4. Check the client version and the logs for the exact error.

Fix

- Fix credentials or reset the password.
- Update or reinstall the VPN client.
- Allow the VPN in the firewall.
- Contact the network admin if the server side is down.

Commands

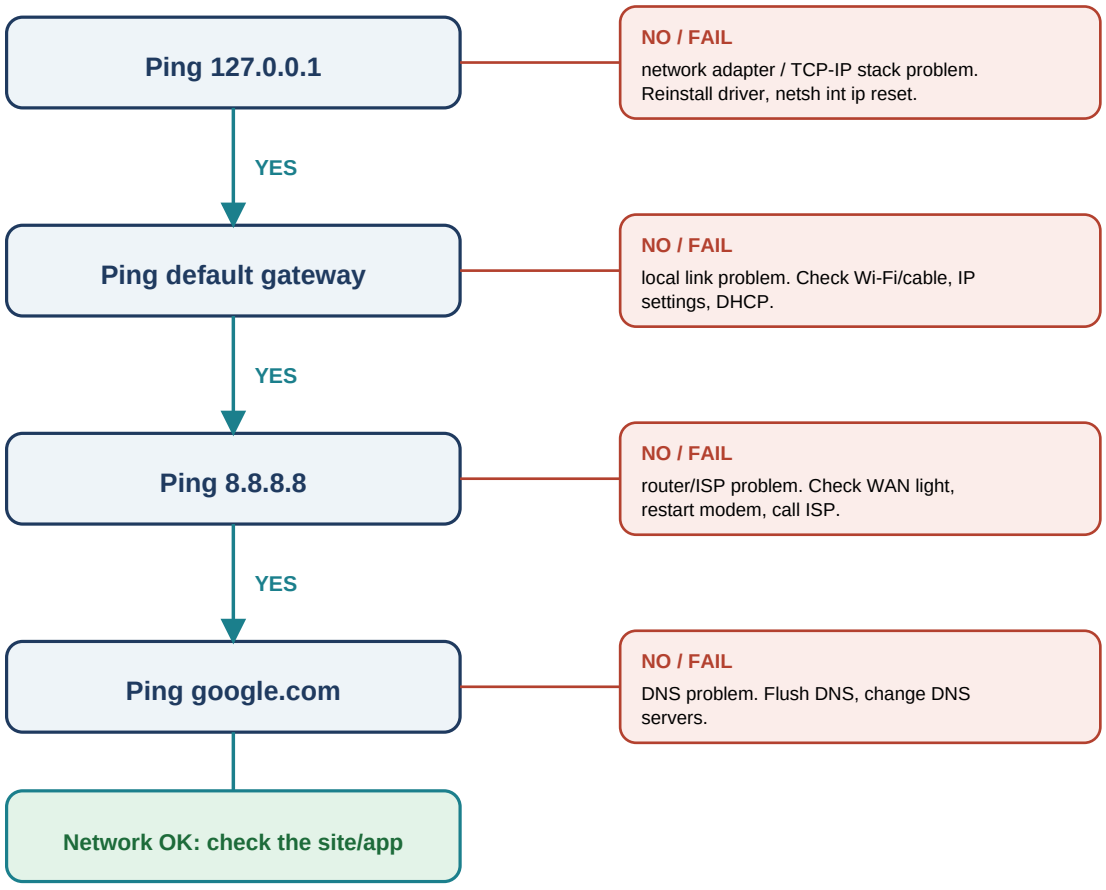
```
ping 8.8.8.8
nslookup <vpn server name>
tracert <vpn server>
```

4. Command cheat sheet

Windows	Linux	Purpose
`ping <host>`	`ping <host>`	Test whether a host replies
`ipconfig /all`	`ip a / ifconfig`	Show IP, mask, gateway, DNS, MAC
`ipconfig /release, /renew`	`sudo dhclient -r / sudo dhclient`	Get a new IP from DHCP
`ipconfig /flushdns`	`resolvectl flush-caches`	Clear the DNS cache
`nslookup <name>`	`dig <name>`	Test DNS resolution
`tracert <host>`	`traceroute <host>`	Show the path and where it breaks
`netstat -an`	`ss -tuln`	Show open ports and connections
`arp -a`	`ip neigh`	Show IP-to-MAC mappings on the LAN
`route print`	`ip route`	Show the routing table and gateway
`netsh wlan show interfaces`	`iwconfig / nmcli dev wifi`	Wi-Fi signal, channel, SSID

5. Troubleshooting flowchart: the ping ladder

Work from top to bottom. The first test that fails shows which layer to fix.



6. Router basics

- Find the router address: it is the default gateway in ipconfig /all (often 192.168.1.1 or 192.168.0.1).
- Open it in a browser and log in. Change the default admin password straight away.
- Change the Wi-Fi name (SSID) and set a strong WPA2 or WPA3 password.
- Check the DHCP range and keep static devices outside it.
- Choose a less crowded Wi-Fi channel (1, 6 or 11 for 2.4 GHz).
- Update the firmware regularly; back up the settings before a factory reset.

7. Automated check (Python)

The included script network_check.py runs the ping ladder automatically and prints a diagnosis. It uses only the Python standard library.

```
python network_check.py
=== Network Check ===

  [PASS] Loopback (127.0.0.1)
  [PASS] Default gateway (192.168.1.1)
  [PASS] Public IP (8.8.8.8)
  [FAIL] DNS lookup (google.com)

--- Diagnosis ---
Internet works but names do not resolve. This is a DNS problem.
```

8. Screenshots

Add your own screenshots here (ipconfig /all, ping tests, tracert, script output) to show the guide was tested on a real machine.